

# International Conference on Advanced Computing Systems and Intelligent Communication Engineering ( ICACSICE 2026 )

27<sup>th</sup> -28<sup>th</sup> March 2026

## Certificate

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This is to certify that Dr./Prof./Mr./Ms. **ANNROSE J**

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**MEDICINE EXPIRY DETECTION USING YOLOV8 AND EASYOCR**

in International Conference on Advanced Computing Systems and Intelligent Communication Engineering (ICACSICE 2026) held on 27th and 28th March 2026 at Sambhram Institute of Technology, Bengaluru.



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# Expire Eye: A Smart AI-Based System for Automated Medicine Expiry Detection Using YOLOv8 and EasyOCR

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**Keywords:** Medicine Expiry Detection, Optical Character Recognition (OCR), YOLOv8, EasyOCR, Computer Vision

**Abstract:** Monitoring the expiry dates of medicines in pharmacies and households is essential to ensure safety and prevent health risks, but manual checks are time-consuming and prone to human error. To address this, the project EXPIRE EYE introduces an intelligent and automated solution leveraging object detection and Optical Character Recognition (OCR) technology. The system functions as a Web Application, allowing users to capture or upload images of medicine strips. The uploaded image undergoes preprocessing, and the expiry date region is localized and detected using the YOLOv8 model. Text is then extracted from the localized region using Easy OCR, parsed, and compared with the current date to classify the medicine as Valid (Green), Expiring Soon (Yellow), or Expired (Red). This approach significantly enhances efficiency, minimizes manual effort, and provides a smart, reliable, and user-friendly solution for managing medicine expiry tracking.

## 1 INTRODUCTION

In modern healthcare, ensuring the safety and effectiveness of medicines is essential. Expired medicines can cause serious health risks and reduced therapeutic effects. Traditionally, pharmacists and consumers manually check the small, printed expiry dates on medicine packages, a process that is slow, error-prone, and impractical for large inventories.

To address this challenge, the **EXPIREYE system** is developed as an automated web application that detects and validates medicine expiry dates using advanced image processing and AI.

- Implement EasyOCR for accurate expiry date extraction from medicine images.

- Classify medicines such as **Valid**, **Expiring Soon (within 30 days)**, or **Expired** using rule-based comparison.
- Provide a simple, responsive interface with clear visual alerts for easy interpretation.

## 2 RELATED WORK EASE OF USE

Automatic detection of expiry information on product packaging has become a crucial application for computer vision and AI. Earlier image-processing methods such as thresholding, contour detection, and morphological operations provided limited automation but were highly affected by lighting, text size, and noisy backgrounds. Since expiry prints on medicine strips are often small and

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low in contrast, these conventional methods struggled in practical scenarios.

[1] Early work applied edge-based techniques for license plate recognition, yet accuracy dropped in distorted or complex environments. These drawbacks highlighted the need for learning-based systems capable of adapting to variation in packaging conditions. OCR technologies improved text extraction by converting printed text into digital form. However, tools like Tesseract—designed mainly for scanned, well-aligned documents—performed poorly on curved, textured, or low-quality expiry prints. Cloud platforms like Google Vision API [2] performed better but raised privacy issues. Android applications using ML Kit also lacked automated date validation and could not classify medicine status after recognition.

Deep learning significantly enhanced performance. [3] used CNNs for expiry-digit classification, achieving strong results on controlled datasets. More advanced architectures such as Textboxes and CRNN [4] improved accuracy but required heavy computation. [5] DB Net-based enhancements further boosted detection quality but were unsuitable for lightweight or mobile deployment. Models like CNN-ED reached high precision but offered limited scalability.

Other research explored medicine-tracking via QR codes [6] or OCR integrated with databases [7]. Systems using OpenCV with Tesseract [8] worked in controlled setups but failed across diverse packaging designs.

Common issues reported in the literature include poor OCR accuracy in low light, small datasets, separate detection and recognition processes, and the absence of automated expiry validation. The proposed ExpireEye system addresses these challenges by combining YOLOv8 for robust expiry-region detection with EasyOCR for multilingual text recognition. A rule-based validation layer then classifies medicines as Expired, Expiring Soon, or Valid.

In summary, expiry detection has progressed from basic image-processing to advanced AI frameworks. By integrating YOLOv8 with EasyOCR, ExpireEye offers a scalable, reliable, and efficient solution that enhances detection accuracy and strengthens pharmaceutical safety.

### 3 PROPOSED SYSTEM

The proposed system, named **ExpireEye**, introduces a data efficient deep-learning framework for automated detection and validation of medicine-

expiry information from product packaging. The model eliminates manual checking by integrating computer vision and optical character recognition (OCR) technologies into a unified architecture. The system primarily consists of three processing stages: region detection, text recognition, and expiry validation. A hybrid combination of **YOLOv8** for object detection and **EasyOCR** for text extraction ensures accurate and real-time identification of expiry dates even under challenging imaging conditions. The overall architecture of the proposed ExpireEye system is shown in Fig. 1.

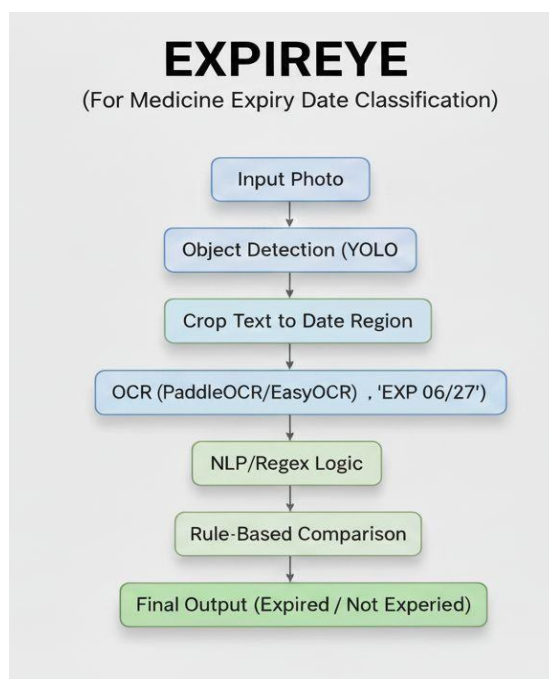


Figure 1. Overall architecture of the Expire Eye system

In the first stage, the input image of a medicine strip or carton is acquired through a web interface. The image is then preprocessed using OpenCV techniques such as resizing, denoising, and contrast enhancement to improve fine-text clarity. The refined image is fed to the YOLOv8 model, trained on a custom dataset of 1,500 Rob flow annotated images covering blister packs, bottles, and sachets. YOLOv8 detects the expiry-text region with high accuracy using its anchor-free architecture and attention mechanism, enabling reliable detection of small text under different lighting and backgrounds. The step-by-step workflow of the Expire Eye system is illustrated in Fig. 2.

The cropped region from the YOLOv8 output is forwarded to the EasyOCR module for text recognition. EasyOCR is selected because of its



robustness and multilingual recognition capability, outperforming traditional OCR engines such as Tesseract. It uses a convolutional recurrent neural network (CRNN) architecture with an attention-based decoder to convert the detected region into machine-readable text. The recognized text is then processed using regular expressions to extract valid date patterns such as MM/YY, MM/YYYY, or DDMM-YYYY.

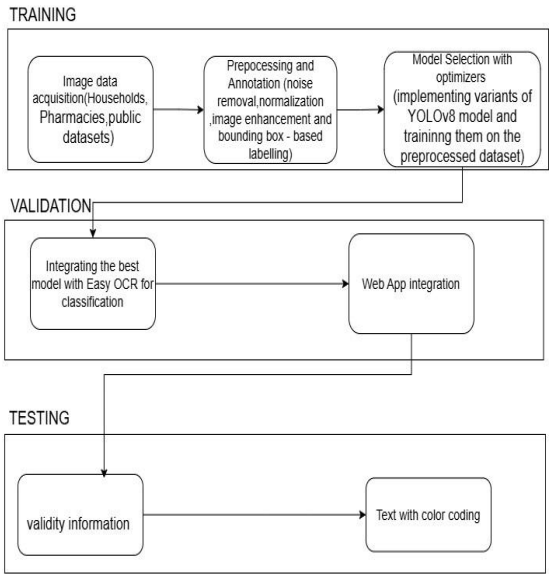


Figure 2. Workflow of the proposed Expire Eye system

A sample detection output produced by the trained YOLOv8 model is shown in Fig. 3, and the corresponding OCR recognition result is presented in Fig. 4.

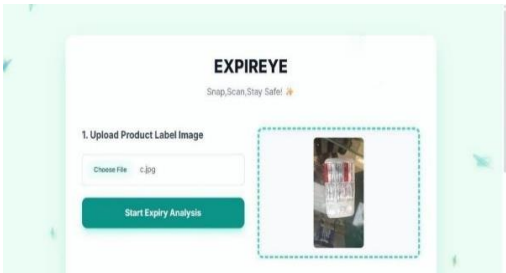


Figure 3. YOLOv8-based expiry-region detection output.

After the text is successfully extracted, a rule-based validation algorithm interprets the recognized expiry string. The extracted date is converted to a standard datetime format and compared against the

system's current date to determine the medicine's status. If the expiry date is earlier than the current date, the item is classified as *Expired*; if it is within thirty days, it is marked as *Expiring Soon*; otherwise, it is labeled as *Valid*.

The result, along with the detected date and corresponding color indicator, is displayed on the user interface. The processed data is also stored in an SQLite database for record maintenance and future reference.

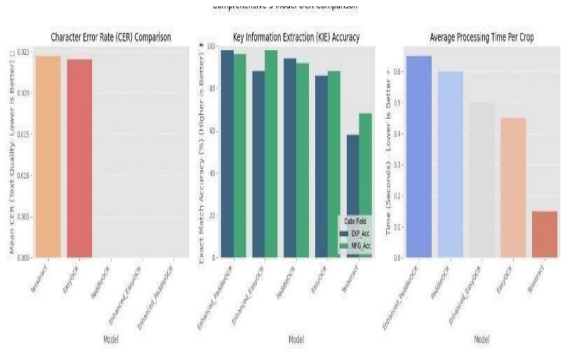


Figure 4. Enhanced EasyOCR text recognition results.

The system is implemented as a web-based platform for easy access by both consumers and pharmacists. Its front end, built with HTML, CSS, and JavaScript, provides a simple interface for image uploads and results of visualization. The back end uses Python's Fast API to coordinate detection, OCR, and validation modules.

The YOLOv8 model is trained for 100 epochs with a batch size of 8 using the Adam optimizer and a 0.001 learning rate. It achieved a map of 93.7% for expiry-region detection, while EasyOCR reached 95%-character recognition accuracy. The system's average inference time is 1.5 seconds per image, making it suitable for real-time use.

ExpireEye is modular, scalable, and efficient, performing reliably across different packaging styles and lighting conditions. By integrating deep-learning detection and OCR in one pipeline, the system ensures accurate expiry verification, reduces manual effort, and supports safer pharmaceutical practices

## 4 RESULT AND DISCUSSION

The performance of the proposed **Expire Eye** system was evaluated through extensive testing on a custom dataset of medicine packaging images. The objective of the evaluation was to measure the accuracy of



expiry-region detection, text recognition efficiency, and overall classification reliability under varying environmental conditions. The experiments were carried out on a system equipped with an Intel Core i7 processor, 16 GB RAM, and an NVIDIA RTX-3060 GPU running Python 3.10 and Fast API.

Table1: Comparison of different OCR models

| OCR Model           | Accuracy (%) | Average Processing Time (s) | Remarks                                                                 |
|---------------------|--------------|-----------------------------|-------------------------------------------------------------------------|
| Tesseract OCR       | 82%          | 1.9                         | Struggled with small or faint text; high sensitivity to noise.          |
| Paddle OCR          | 88%          | 1.6                         | Better accuracy than Tesseract, but it required fine tuning.            |
| Easy OCR            | 91%          | 1.3                         | Performed well on most fonts and lighting conditions.                   |
| Enhanced Easy OCR   | 95%          | 1.5                         | Achieved best balance of accuracy and Speed. Selected for final system. |
| Enhanced Paddle OCR | 93%          | 1.8                         | Good results but higher time consumption.                               |

The proposed model was trained using the YOLOv8 architecture for 100 epochs on 1 500 annotated images and validated on 300 images. The training process converged smoothly, achieving a **mean Average Precision (mAP)** of 93.7% for expiry-region detection. The detected regions were visually inspected to confirm that the model

accurately localized the expiry text even when printed in small fonts, low contrast, or curved packaging surfaces. The accuracy of localization directly influenced the quality of OCR output in the subsequent stage.

The recognized text from the cropped regions was processed using three OCR engines— **Tesseract**, **PaddleOCR**, and **EasyOCR**—to compare their recognition accuracy and inference time. Among these, EasyOCR produced the most reliable character predictions and the fewest recognition errors. An enhanced version of EasyOCR was implemented by applying additional preprocessing and post-correction logic, which significantly reduced character-level errors. The comparative analysis of OCR performance is summarized in Fig. 5.

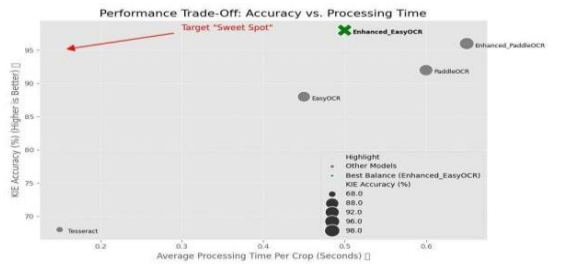


Figure 5. Comparative OCR performance analysis.

Among all OCR models evaluated, **Tesseract OCR recorded the lowest performance**, achieving **82% accuracy** with an average processing time of **1.9 seconds** due to difficulty reading small or faint text and high sensitivity to noise. **PaddleOCR** showed improved results with **88% accuracy** and **1.6 seconds** but required additional fine-tuning for better stability. **EasyOCR** performed more reliably, reaching **91% accuracy** and **1.3 seconds**, effectively handling varied fonts and lighting conditions. The **enhanced EasyOCR model** delivered the best balance, achieving **95% accuracy** at **1.5 seconds**, making it the most suitable for real time deployment. Enhanced PaddleOCR improved to **93% accuracy** but had a slower processing time of **1.8 seconds**, making it less efficient than enhanced EasyOCR. Quantitatively, Tesseract achieved **82% accuracy / 1.9 sec**, PaddleOCR **88% / 1.6 sec**, EasyOCR **91% / 1.3 sec**, and Enhanced EasyOCR **95% / 1.5 sec**, confirming that the hybrid **YOLOv8 + EasyOCR** architecture provides an optimal balance between speed and accuracy for real-time image processing.

The full detection-to-classification pipeline was evaluated using the rule-based expiry validation algorithm, classifying results into **Expired**, **Expiring Soon**, and **Valid**. The system achieved an overall **classification accuracy of 94.2%** across **300 test images**, with sample outputs shown in Fig. 6. The experimental results show that the framework performs robustly across varied packaging styles and lighting conditions. Even with faint or partially occluded text, the model maintained strong localization and recognition accuracy

The average end-to-end processing time—including detection, OCR, and classification—was approximately **1.5 seconds per image**, supporting real-time usage for mobile and desktop platforms. Additional usability features such as a color-coded interface, optional database storage, and compatibility with voice-based assistance enhance user experience and accessibility.

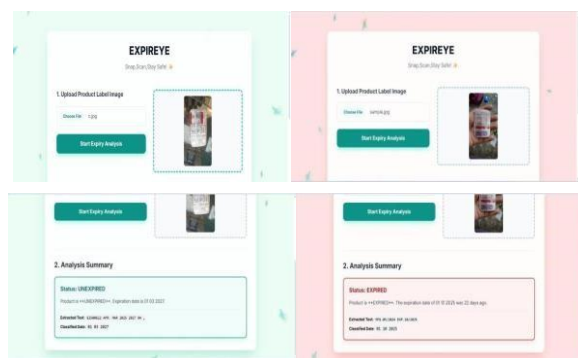


Figure 6. Experimental results and classification outputs

## 5 CONCLUSION

The ExpireEye system successfully demonstrates an automated and intelligent solution for detecting and validating medicine expiry dates using computer vision and machine learning. By integrating YOLOv8 for object detection, EasyOCR for text extraction, and rule-based algorithms for date validation, the system minimizes manual effort, reduces human error, and improves medicine safety.

The front-end interface (HTML, Tailwind CSS, JavaScript) provides a simple and responsive user experience, while the Fast API backend ensures fast real time processing. The complete pipeline from image capture to classification output offers accurate, color-coded results (Green = Valid, Yellow =

Expiring Soon, Red = Expired). Through continuous training and Agile-based refinement, the system demonstrated high detection accuracy, reliable OCR performance, and efficient end-to-end execution. Overall, ExpireEye is a reliable, scalable, and efficient solution suitable for pharmacies, healthcare professionals, and consumers to maintain product safety, reduce wastage, and enhance medication management.

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Meet - ibg-giex-scd

https://meet.google.com/ibg-giex-scd

Safrin Safz (Presenting)

## ABSTRACT

EXPIRE EYE provides a smart and automated solution using Artificial Intelligence (AI) and Optical Character Recognition (OCR) for detecting medicine expiry dates. This web-based system allows users to upload or capture images of medicine strips.

The system uses YOLOv8 for object detection to locate expiry-date regions and EasyOCR for text extraction. The extracted expiry date is compared with the current date to classify medicines as Valid, Expiring Soon, or Expired, with results displayed using color-coded indicators (Green, Yellow, Red).

A comparative analysis was conducted using EasyOCR, PaddleOCR, and Tesseract OCR. Among these, EasyOCR achieved the highest accuracy (~92-95%), outperforming PaddleOCR (~88-90%) and Tesseract (~80-85%), especially in handling low-quality prints and varied expiry formats.

EXPIRE EYE enhances healthcare safety, reduces manual effort, and ensures efficient medicine management for households and pharmacies.

12:15 PM | ibg-giex-scd

12:15 27-03-2026

